

MLOps Assignment 2

Fine-Tuning DistilBERT for Book Genre Classification

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1. Overview

This report documents an end-to-end MLOps pipeline for fine-tuning a pre-trained transformer model on the UCSD Goodreads book review dataset. The pipeline covers model loading from Hugging Face Hub, experiment tracking with Weights & Biases (W&B) on Kaggle, model evaluation, and deployment back to Hugging Face Hub.

2. Model Selection & Justification

The model selected for this assignment is distilbert-base-uncased from Hugging Face Hub. DistilBERT is a distilled version of BERT that retains 97% of language understanding while being 40% smaller and 60% faster. This makes it ideal for Kaggle's free GPU tier where compute hours are limited. The model was pre-trained on a large English corpus, making it well-suited for classifying Goodreads book reviews across 8 genres. Its compatibility with the HuggingFace Trainer API also simplified the MLOps pipeline setup considerably.

3. Pipeline Setup & Training

The Kaggle notebook was configured with T4 x2 GPU, Internet enabled, and API keys stored securely via Kaggle Secrets. The dataset comprised 2,000 reviews per genre, split 80/20 into train and test sets (6,400 train / 1,600 test).

Training Parameter	Value	Training Parameter	Value
Epochs	3	Learning Rate	5e-5
Train Batch Size	10	Warmup Steps	100
Eval Batch Size	16	Weight Decay	0.01
Eval Strategy	steps	report_to	wandb

The `compute_metrics` function was extended to track accuracy, weighted F1, precision, and recall at each evaluation step. All metrics were logged to the W&B dashboard in real time.

4. Evaluation Results

Metric	Score
Accuracy	61.13%
F1 Score (Weighted)	61.19%
Precision (Weighted)	61.37%
Recall (Weighted)	61.13%
Eval Loss	2.315

The model achieved 61.13% accuracy on the 8-class genre classification task, significantly above the random baseline of 12.5%. Balanced precision and recall scores indicate the model generalises well across genres. Some misclassifications are expected due to overlapping vocabulary between genres such as fantasy and young adult.

5. Challenges & Solutions

Challenge	Solution
W&B not initialized error on <code>evaluate()</code>	Removed early <code>wandb.finish()</code> ; kept single <code>wandb.init()</code> open throughout
W&B logging disabled in starter notebook	Removed <code>WANDB_DISABLED</code> env variable; set <code>report_to="wandb"</code>
Secure API key management	Used Kaggle Secrets (<code>UserSecretsClient</code>) for <code>WANDB_API_KEY</code> and <code>HF_TOKEN</code>

6. Artifacts & Submission Links

- Fine-tuned model and tokenizer pushed to Hugging Face Hub
- Classification report saved and uploaded as W&B Artifact
- Kaggle notebook made public; GitHub repository created with notebook and README

Kaggle Notebook: <https://www.kaggle.com/code/dhruvig25ait2030/mlops-assignment2>

Hugging Face Model: <https://huggingface.co/Doobieeeful/distilbert-genre-classification>

W&B Dashboard: <https://wandb.ai/g25ait2030-prom-iit-rajasthan/mlops-assignment2>

GitHub Repository: <https://github.com/dhruvi9660/MLOpsassignment2/>
